

Akhileshwar Pratap Singh

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EDUCATION

Rajiv Gandhi Institute of Petroleum Technology <i>B.Tech in Electronics Engineering (CGPA: 7.69/10)</i>	Jais, Amethi 2023 – 2027
City Montessori School, Mahanagar <i>ISC Board (92.5%)</i>	Lucknow 2021 – 2022
City Montessori School, Mahanagar <i>ICSE Board (91.2%)</i>	Lucknow 2019 – 2020

EXPERIENCE

Bharat Electronics Ltd.  <i>Embedded Systems Intern</i>	Pune May 2025 – July 2025
- Designed power adapter PCB for Laser Range Finder device used in Indian defense applications - Integrated ROS2 framework packages for robotics navigation and autonomous control systems	
Yapassio Pvt. Ltd.  <i>Research and Development Intern</i>	Remote May 2024 – July 2024
- Improved system efficiency by 40% and reduced operational expenses by 39% - Engineered 12 prototypes leading to 6 successful feature launches	

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, CMake
Robotics Stack: ROS2 (**Humble**), RViz2, Linus(Ubuntu) Gazebo, Fusion 360, Navigation2, SLAM, URDF
Tools: Arduino IDE, TinkerCAD, KiCAD, EasyEDA, Oscilloscope, Git
Frameworks: ROS2, Arduino, LoRa, Bluetooth Communication

PROJECTS

Autonomous Mobile Robot with SLAM and Navigation <i>ROS2, Python, Gazebo, RViz2</i>  - Designed and implemented a differential drive autonomous mobile robot using ROS2 Humble with a modular software architecture - Integrated Cartographer SLAM for real time 2D mapping and localization using LiDAR data - Integrated Nav2 based autonomous navigation including path planning, obstacle avoidance and goal execution - Worked extensively with ROS2 nodes, topics, services, TF frames and launch files for system orchestration - Validated robot behavior in Gazebo simulation and visualized sensor data, maps and trajectories using RViz2	Dec 2025 – Jan 2026
Differential Drive Mobile Robot–Hardware Modeling & URDF <i>Fusion 360, ROS2, URDF/Xacro, Gazebo</i>  - Developed a differential-drive mobile robot in Fusion 360 including chassis, wheels, motors, and sensor mounts - Converted the mechanical model into a ROS2-compatible URDF/Xacro with correct link–joint structure - Defined the complete TF tree (base_footprint, base_link, wheels, sensors) for navigation and localization - Configured wheel revolute joints, collision/visual meshes, and validated rendering in RViz2 and Gazebo	Jan 2026
Industrial Biogas Plant Automation System <i>Industrial Relays, Timer Relay, Solenoid Valves</i>  - Worked on fail-safe mechanisms for industrial automation ensuring 99.9% uptime - Engineered microcontroller-free control system with three-tier safety alerts (Patent Pending)	Dec 2025 – Jan 2026
Custom Carrier Board <i>KiCAD, Arduino, Motor Drivers</i>  - Formulated PCB reducing wiring complexity by 88% and assembly time by 50% - Optimized power distribution and signal integrity for robotic applications	Dec 2024 – Jan 2025

ACHIEVEMENTS

Winner - Urjotsav Constructed autonomous maze-solving robot outperforming 280+ teams	2024
1st Runner-Up - Meshmerize (IIT Bombay TechFest) Secured podium finish among 2000+ teams in national competition	2024
DRDO Physics Olympiad Among the top 50 students selected nationally	2021

LEADERSHIP

Head - IEEE Robotics and Automation Society, RGIPT - Leading 50+ member society organizing workshops and robotics competitions Speaker at RGIPT X IIT Bombay Techfest Robotics Workshop with 240+ students participation	Nov 2024 – Present
Event Coordinator - Science and Technical Council, RGIPT Coordinated technical events and festivals with multi-institutional participation	2024